Gauge Structure and Effective Geometry from Admissible Non-Injective Projection

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Abstract

Starting from the axiomatic foundation of admissible non-injective transitions [1], we derive the macroscopic structures of gauge theory and effective spacetime geometry without introducing them as postulates. The derivation proceeds in three steps. First, gauge charges are identified as invariants of equivalence classes in the projection fibre: two configurations are gauge-equivalent when they project to the same observable state. The group of fibre-preserving transformations constitutes the gauge group, with U(1), SU(2), and SU(3) identified as natural candidates from the admissibility structure. Second, in the continuum limit $q \to \infty$ established in [1], the discrete admissibility filter Π_n converges to an effective continuum operator $\mathcal{L}_\Pi$ whose principal symbol defines an effective metric tensor $g^{\mu\nu}(x) \propto A^{\mu\nu}(x)$. Third, the bounded Born–Infeld admissibility constraint (A3–A4) restricts the admissible forms of $A^{\mu\nu}(x)$: in homogeneous regimes, hyperbolicity and spectral stability uniquely select Minkowski geometry; in the presence of a localized stationary obstruction, flux conservation forces the Schwarzschild geometry as the unique stationary exterior solution. The gauge connection arises as the correction required to compare admissible fibres over neighbouring points, and Yang–Mills dynamics follow from minimising the curvature of this connection. No dynamical substrate is required at any step; the derivations rest solely on Axioms A1–A4 of [1] and the spectral results of the O-series.

1 Introduction

Position in the programme

The companion papers [1, 2, 3] establish the quantum-mechanical structure of the Cosmochrony framework: the proto-state, phase coherence, the Born rule, the singlet correlator, and the Tsirelson bound, all derived from the admissibility structure of non-injective projection. These results concern the *internal* structure of the admissible fibre at a single transition step.

The present paper addresses the *macroscopic* consequences of allowing the projection structure to vary from point to point in the effective continuum limit. Two questions drive the paper:

(i) What are the conserved invariants of the projection fibre, and what symmetry group do they generate?

(ii) What effective geometry is selected when the admissibility constraints of [1] are imposed on a spatially varying projection?

The answers are: gauge charges and gauge groups (§3), and Lorentzian geometry with Schwarzschild as the exterior stationary solution (§6).

What is and is not derived in this paper

This paper derives the *structure* of gauge theory and effective geometry from the admissibility framework. It does not derive specific coupling constants, the precise matter content of the Standard Model, or the quantitative value of Newton's constant. These remain open and are identified as directions for future work in Section 7.

The derivations are structural: they identify which gauge groups and which effective geometries are *admissible* given the constraints, not which specific values their parameters take.

Organisation

Section 2 establishes the continuum limit of the discrete projection Π_n and the emergence of an effective differential operator. Section 3 derives gauge charges as fibre invariants and identifies the gauge group. Section 4 derives the gauge connection from the requirement of consistent transport between neighbouring fibres. Section 5 derives Yang–Mills dynamics from the minimisation of connection curvature. Section 6 derives the effective metric from the principal symbol of $\mathcal{L}_\Pi$ and establishes the emergence of Minkowski and Schwarzschild geometries. Section 7 identifies the open directions.

2 Continuum Limit of the Admissibility Projection

2.1 From discrete Π_n to a continuum operator

The discrete projection Π_n acts on the orbit span $\Omega_n^{(c)} \subset \mathbb{C}^q$, filtering admissible directions via the Born–Infeld bound $A_n \leq c_\chi/\sqrt{\lambda_n}$ (Foundation M,

§**??**). As established in [1] Section 3.5, the large-q limit sends:

$$G_q \longrightarrow \mathrm{Heis}_3(\mathbb{R}), \tag{1}$$

$$\mathbb{C}^q \longrightarrow L^2(\mathbb{R}), \tag{2}$$

$$\text{BFS shells } S_n \longrightarrow \text{continuous foliation}, \tag{3}$$

$$\Pi_n \text{ (discrete)} \longrightarrow \mathcal{L}_\Pi \text{ (continuum operator)}. \tag{4}$$

The continuum limit of relational Laplacians on graphs of the type $\mathrm{Cay}(G_q, S_q)$ is governed by standard spectral convergence results [5]: under mild regularity conditions, the graph Laplacian converges in the strong-resolvent sense to a second-order differential operator

$$\mathcal{L}_\Pi \;=\; \nabla_\mu\big(A^{\mu\nu}(x)\,\nabla_\nu\big), \tag{5}$$

where $A^{\mu\nu}(x)$ is a symmetric tensor encoding the local connectivity and stiffness of the admissible projection.

Remark 2.1 (Status of the continuum limit)**.** *The convergence $\Pi_n \to \mathcal{L}_\Pi$ is established qualitatively in [1] and in detail in [5] for the class of weighted relational Laplacians. A fully rigorous proof at the level of the Heisenberg Cayley graph $\mathrm{Cay}(G_q, S_q)$ as $q \to \infty$ remains open. The present section uses the continuum limit as a working hypothesis, stated honestly as such.*

2.2 The operator-metric correspondence

In admissible continuum regimes, the operator $\mathcal{L}_\Pi$ has a well-defined principal symbol:

$$\sigma_2(\mathcal{L}_\Pi)(x,k) \;=\; A^{\mu\nu}(x)\,k_\mu k_\nu. \tag{6}$$

This symbol defines an effective metric tensor [5]:

$$g^{\mu\nu}(x) \;\propto\; A^{\mu\nu}(x). \tag{7}$$

The metric is not an independent dynamical variable: it is a derived object encoding how admissible modes propagate in the continuum limit. This is the operator-metric correspondence on which all subsequent geometric results rest.

3 Gauge Charges as Fibre Invariants

3.1 Charges from non-injectivity

The non-injectivity of Π_n (Axiom A2 of [1]) means that multiple configurations in $\Omega_n^{(c)}$ project to the same observable state O_n. These configurations are physically indistinguishable at the observable level but may carry distinct internal structure.

Definition 3.1 (Gauge charge)**.** *A* gauge charge *is a functional*

$$Q : \Omega_n^{(c)} \longrightarrow \mathbb{R} \tag{8}$$

that is constant on projection fibres:

$$\Pi_n(\omega_1) = \Pi_n(\omega_2) \implies Q(\omega_1) = Q(\omega_2). \tag{9}$$

A gauge charge is therefore an observable that cannot be resolved by Π_n itself — it lives in the kernel structure of Π_n — but is well-defined on the fibre.

Definition 3.2 (Gauge group)**.** *The* gauge group $\mathcal{G}$ *is the group of transformations $\omega \mapsto g \cdot \omega$ that preserve the projection:*

$$\mathcal{G} := \big\{ g \in \mathrm{Aut}(\Omega_n^{(c)}) \mid \Pi_n(g \cdot \omega) = \Pi_n(\omega) \text{ for all } \omega \big\}. \tag{10}$$

$\mathcal{G}$ consists precisely of the fibre-preserving symmetries: it acts within $\Pi_n^{-1}(O_n)$ without affecting the observable output.

Remark 3.3 (Gauge invariance as structural necessity)**.** *The gauge group is not introduced by hand. It is forced by the structure of Π_n: as soon as Π_n is non-injective, there exist non-trivial fibre-preserving transformations, and hence a non-trivial gauge group. Gauge invariance is a structural consequence of Axiom A2, not an additional postulate.*

3.2 Identification of the gauge groups

Three natural candidates emerge from the fibre structure established in Q1 [2] and Q2 [3]:

- U(1): global phase rotations $\omega \mapsto e^{i\theta}\omega$ within the complex fibre $\mathbb{C}^q$. These preserve $|\omega|$ and hence the observable norm, while acting non-trivially on the phase — which is in the kernel of Π_n by the phase invariance of $\sigma_c(n)$ (O12).
- SU(2): the stable fixed point of the admissibility flow established in [3] Theorem 8.1. Its fibre-preserving action on $\mathbb{C}^2$ is the spin-$\frac{1}{2}$ structure derived in [2].
- SU(3): a structurally motivated candidate for a higher-dimensional internal symmetry of the fibre, corresponding to multiple co-admissible internal directions beyond SU(2). Its identification from the O-series admissibility data is an open direction (Section 7).

Remark 3.4 (Status of SU(3))**.** *The identification of* SU(3) *as a gauge group is a structural hypothesis, not a derived result. Establishing it would require finding, within the O-series spectral data, a co-admissibility pattern for a group with* SU(3) *structure analogous to the $\lambda_{1/2} = \lambda_{3/2}$ co-admissibility of [3]. This is identified as a medium-term open direction.*

4 The Gauge Connection

4.1 The problem of comparing neighbouring fibres

In the continuum limit, the admissible fibre F_n becomes a smooth fibre bundle over the effective spacetime manifold. For each point x in this manifold, the fibre $F_x = \Pi^{-1}(x)$ carries the internal gauge structure of Section 3.

When the projection $\Pi(x)$ varies from point to point, the fibres F_x and F_{x+dx} are not canonically identified. A state $\psi(x)$ transforming under $\mathcal{G}$ acquires an additional term under differentiation:

$$\partial_\mu(g(x)\,\psi(x)) \;=\; g(x)\,\partial_\mu\psi(x) + (\partial_\mu g(x))\,\psi(x), \tag{11}$$

which is not covariant under local gauge transformations $g(x) \in \mathcal{G}$.

4.2 Gauge connection from fibre transport

Definition 4.1 (Gauge connection)**.** *A* gauge connection A_μ *is the correction required to define a covariant derivative*

$$D_\mu \;:=\; \partial_\mu + A_\mu \tag{12}$$

such that $D_\mu\psi$ transforms homogeneously under local gauge transformations.

Proposition 4.2 (Admissibility interpretation of A_μ)**.** *In the projection framework, $A_\mu(x)$ encodes the local variation of the admissible fibre structure: it is the infinitesimal correction needed to transport an admissible configuration from the fibre at x to the fibre at $x + dx$ consistently with the admissibility constraints.*

Structural argument. The admissibility constraints (A1–A3 of [1]) define, at each point x, a fibre F_x of structurally compatible configurations. When $\Pi(x)$ varies, the fibres F_x and F_{x+dx} are related by the differential of the projection map. Consistency of the admissibility condition across neighbouring fibres — i.e. that a configuration admissible at x maps to an admissible configuration at $x+dx$ — requires a linear correction of the form $A_\mu(x)\,dx^\mu$. This correction is precisely the gauge connection. □

Remark 4.3 (Relation to the holonomy argument of the companion paper)**.** *The white paper [6] establishes that non-injective admissible projections generically have non-trivial holonomy, which is the minimal structural condition for a gauge symmetry to arise as a descriptive invariance. Proposition 4.2 is the local infinitesimal version of this global statement.*

5 Yang–Mills Dynamics

5.1 Field strength as curvature of the connection

Definition 5.1 (Field strength)**.** *The* field strength *associated with the gauge connection A_μ is the curvature of the covariant derivative:*

$$F_{\mu\nu} \;:=\; [D_\mu, D_\nu] \;=\; \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]. \tag{13}$$

$F_{\mu\nu}$ measures the failure of commutativity of parallel transport around an infinitesimal loop — equivalently, the non-commutativity of the admissibility constraints in two neighbouring directions.

5.2 Yang–Mills action from admissibility minimisation

Proposition 5.2 (Yang–Mills action from coherence minimisation)**.** *The natural dynamical principle for the gauge connection is minimisation of the deformation of the projection structure, leading to the action*

$$S_{\mathrm{YM}}[A] \;=\; \int \mathrm{Tr}\big(F_{\mu\nu}F^{\mu\nu}\big)\,\sqrt{-g}\,d^4x. \tag{14}$$

Structural argument. By Proposition 4.2, A_μ encodes the local variation of the fibre structure. The curvature $F_{\mu\nu}$ measures the incompatibility of this variation around closed loops: a flat connection ($F_{\mu\nu} = 0$) corresponds to a globally consistent fibre structure, while non-zero curvature signals inconsistency. The admissibility principle (A3 of [1]) favours configurations that preserve the full multiplicity of open directions without premature selection. Translated to the gauge connection, this principle selects configurations that minimise incompatibility — i.e. minimise $\|F_{\mu\nu}\|^2$. The action (14) is the unique gauge-invariant, Lorentz-invariant, second-order functional of A_μ that achieves this. □

Corollary 5.3 (Yang–Mills equations)**.** *Variation of* (14) *with respect to A_μ yields*

$$D_\mu F^{\mu\nu} \;=\; 0, \tag{15}$$

the Yang–Mills equations in vacuum.

Remark 5.4 (Status)**.** *Proposition 5.2 is a structural motivation, not a complete proof. The uniqueness claim requires showing that* (14) *is the* only *admissibility-compatible action at second order in A_μ. This follows in the Born–Infeld limit from the results of [5] (the BI action reduces to the Yang–Mills action in the weak-field regime), but a direct derivation from the axioms of [1] alone is an open step identified in Section 7.*

6 Effective Spacetime Geometry

6.1 Metric from the principal symbol

The operator-metric correspondence (7) gives the effective metric as

$$g^{\mu\nu}(x) \;\propto\; A^{\mu\nu}(x). \tag{16}$$

This metric is not postulated: it encodes how admissible modes propagate in the continuum limit of the projection.

6.2 Signature selection from admissibility

Theorem 6.1 (Lorentzian signature from spectral admissibility, after [5])**.** *In homogeneous and quasi-isotropic admissible regimes, the Born–Infeld admissibility constraint (A3–A4 of [1]) forces the effective metric to have Lorentzian signature* $(-,+,+,+)$*.*

Proof. The bounded Born–Infeld flux constraint excludes purely diffusive (parabolic) effective descriptions, which would imply instantaneous propagation of perturbations and contradict the finite capacity bound of A4. A Riemannian signature $(+,+,+,+)$ produces an elliptic principal symbol, which is also excluded. An ultra-hyperbolic signature (more than one time direction) fails to admit a globally well-posed initial value problem and is excluded by the spectral stability requirement. The unique remaining option compatible with (i) hyperbolicity of $\sigma_2(\mathcal{L}_\Pi)$, (ii) finite characteristic cone, (iii) the monotonic temporal ordering of Foundation M §4.3, and (iv) the bounded admissibility constraint of A4, is the Lorentzian signature $(-,+,+,+)$. The detailed argument is established in [5] Sections 5–6, which depend only on the bounded-flux constraint and spectral stability — both present in the Foundation M axioms — and not on any dynamical substrate. □

6.3 Schwarzschild geometry from flux conservation

Theorem 6.2 (Schwarzschild from bounded admissibility, after [5])**.** *In the presence of a localised stationary obstruction to the admissibility flow, with spherical symmetry and no additional microscopic scales, the unique exterior stationary solution of the effective geometry induced by* $\mathcal{L}_\Pi$ *is the Schwarzschild metric*

$$ds^2 \;=\; -\Big(1-\frac{r_s}{r}\Big)\,dt^2 + \Big(1-\frac{r_s}{r}\Big)^{-1}\,dr^2 + r^2 d\Omega^2, \tag{17}$$

where r_s *is an integration constant characterising the strength of the obstruction.*

Proof. In the exterior region, the homogeneous equation for the effective operator (5) under spherical symmetry reduces to

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2 A^r(r)\frac{d\Phi}{dr}\right) = 0, \tag{18}$$

expressing conservation of admissible flux through spheres of radius r. The general solution is $\Phi(r) = \Phi_0 - C/r$, giving a $1/r$ decay profile. Via the operator-metric correspondence (7), this profile translates to the metric coefficients $g_{tt} = -f(r)$, $g_{rr} = f(r)^{-1}$ with $f(r) = 1 - r_s/r$. The parameter r_s appears as the integration constant of the flux conservation equation and is identified operationally via its influence on admissible propagation cones. This derivation requires only: (i) the bounded-flux constraint (A3–A4), (ii) local injectivity of the projection in the exterior region, (iii) spherical symmetry and stationarity, (iv) absence of additional microscopic scales. No independent metric field equations or energy-momentum tensor are postulated. The detailed argument is established in [5] Sections 6–8. □

Remark 6.3 (Horizon as projection degeneracy)**.** *At $r = r_s$, the lapse function $f(r_s) = 0$. In the present framework, this corresponds to a degeneracy of the principal symbol $\sigma_2(\mathcal{L}_\Pi)$: the operator loses hyperbolicity and the continuum chart ceases to be projectable. The horizon is therefore not a physical singularity of the underlying admissible structure — it is the breakdown of the effective continuum description. This interpretation is consistent with the non-injectivity result of [4]: at the horizon, distinct microscopic configurations become maximally indistinguishable, maximising the projection entropy S_Π.*

6.4 Einstein equations as consistency conditions

Remark 6.4 (Einstein equations from coherence functional)**.** *The white paper [6] establishes, via the infrared expansion of the coherence functional (Appendix A.10), that the Einstein–Hilbert action arises as the leading two-derivative infrared contribution. The Einstein equations $G_{\mu\nu} = 8\pi T_{\mu\nu}$ therefore emerge as consistency conditions of emergent geometry — not as microscopic laws of the admissible substrate. In the language of the present paper: they express the balance between the curvature of the projection-induced metric (encoding admissibility constraints) and the energy-momentum content (encoding localised obstructions to the admissibility flow).*

7 Open Directions

Rigorous continuum limit. The convergence $\Pi_n \to \mathcal{L}_\Pi$ for the Heisenberg Cayley graph $\mathrm{Cay}(G_q, S_q)$ in the $q \to \infty$ limit is established qualita-

tively in [1]. A fully rigorous proof at the level of the spectral O-series data would close this gap.

Direct derivation of Yang–Mills uniqueness. Proposition 5.2 gives a structural motivation for the Yang–Mills action. Its uniqueness — that S_{YM} is the *only* admissibility-compatible second-order action — follows from the Born–Infeld limit [5] but not yet from Foundation M axioms alone.

SU(3) from admissibility. The identification of SU(3) as a gauge group from the admissibility flow requires finding a co-admissibility pattern for an SU(3)-type group in the O-series spectral data. The analogue of the $\lambda_{1/2} = \lambda_{3/2}$ result on $2I$ (established in [3]) must be found for a group with SU(3) structure.

Coupling constants. The present paper identifies which gauge groups and geometries are admissible. It does not determine the numerical values of coupling constants (g_s, g_w, G_N, α). These are expected to emerge from the quantitative spectral data of the O-series via the transfer chain $c_\chi \to \delta_{\mathrm{pair}} \to \beta^* \approx 0.126$, but the explicit identification is open.

Cosmological implications. In homogeneous regimes, the admissibility flow gives a linear growth of cumulative projected capacity $I(n)$, underlying the emergent Hubble law and effective cosmological expansion [6]. The precise connection between the spectral exponent δ_{pair} and the observable cosmological parameters (H_0, w) is an open quantitative problem.

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